Hankyeol Lee’s Portfolio

I work where the cues are unreliable, and redesign what the model is given.

PAPER 1

REVIVE3D: REtrieval-based Volumetric Infusion via Visual Editing

***Lee, Hankyeol**, Baek, Wooyeol, Kim, Seongdo, and Kim, Jongyoo.*

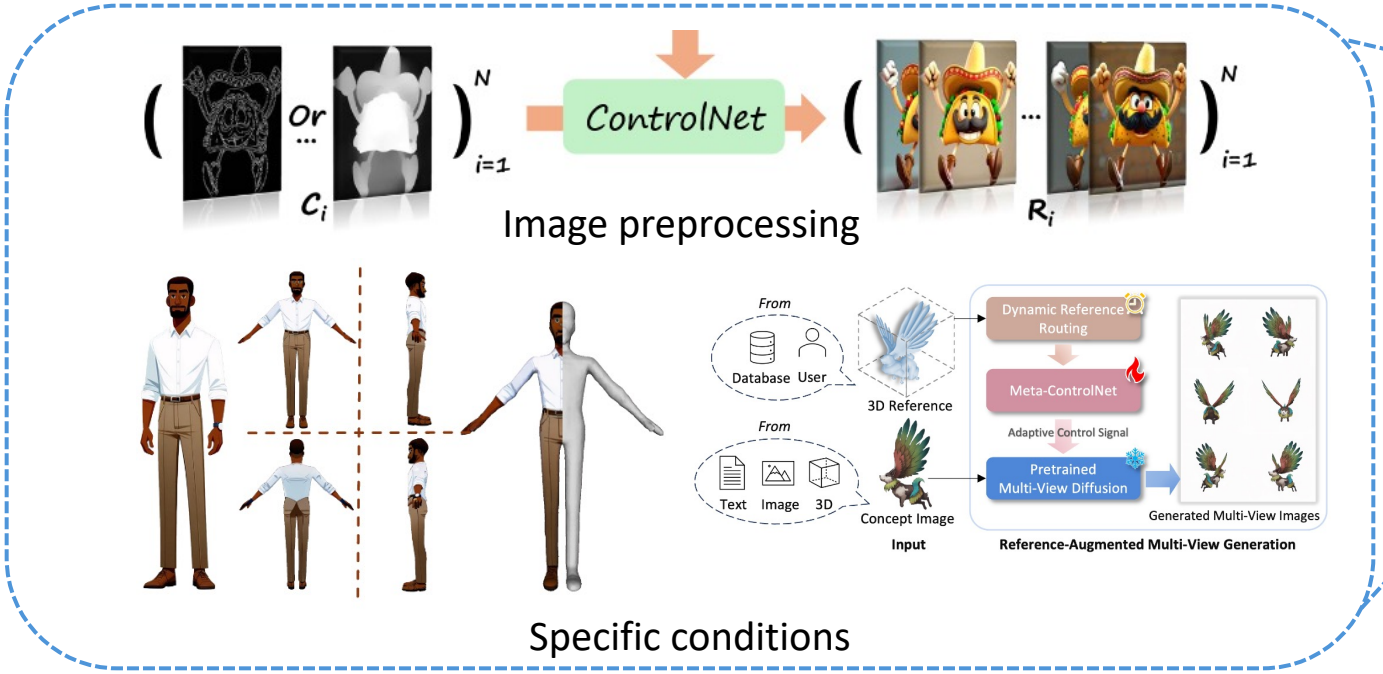
*Accepted at **ICCV Workshop 2025***

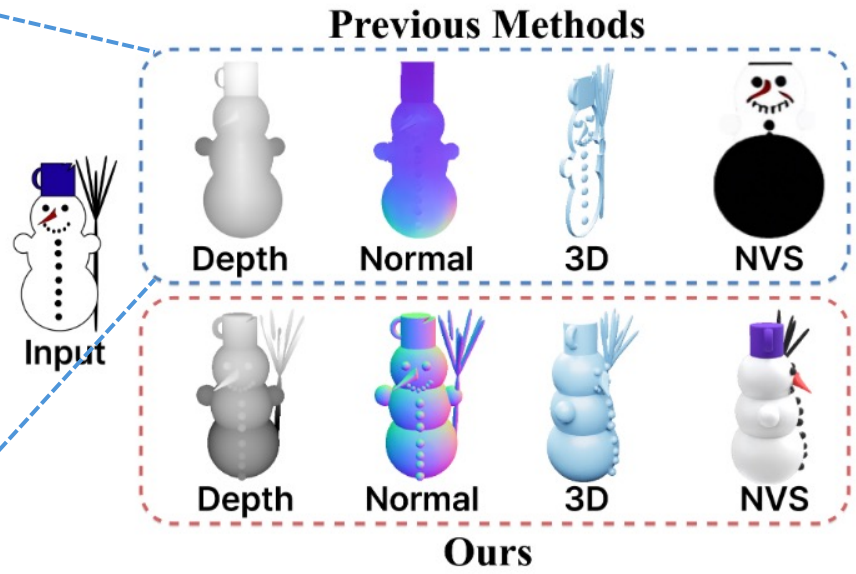
Related Topics

- *3D Generative Model*
- *Retrieve-and-Edit*
- *Cartoon Image to 3D Generation*

Link: https://openreview.net/forum?id=MTAv9V2mmb

MOTIVATION





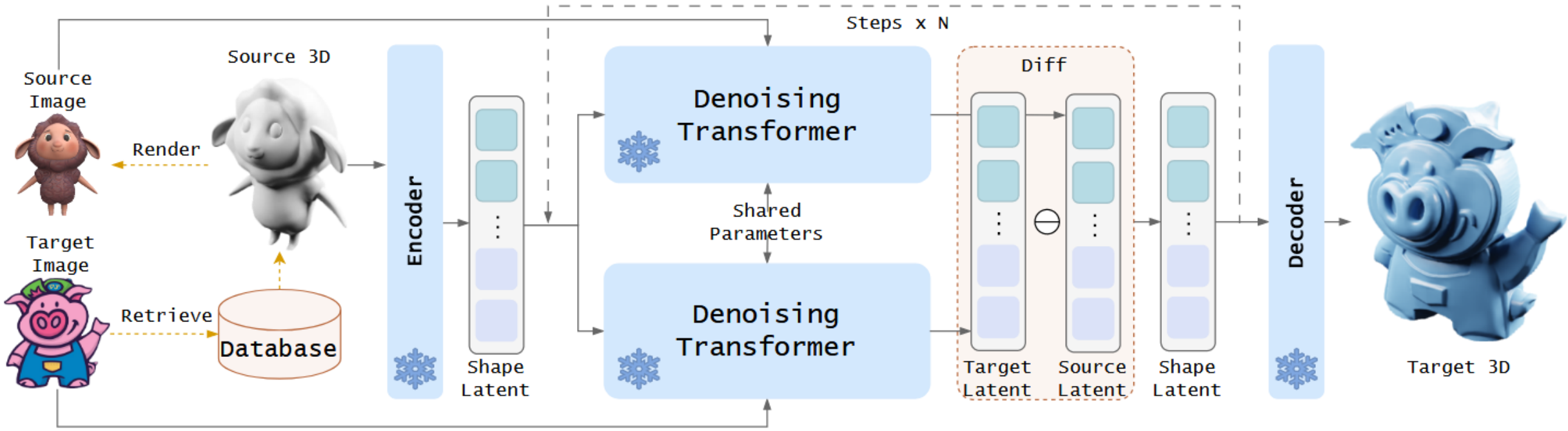
Flat Image

- Existing 3D generative models **fail to generate sufficient volume** when 3D cues are scarce.

Previous Methods

- Pose- or retrieval-conditioned approaches **restrict generation** to specific input conditions.
- Depth/edge-based preprocessing **alters the original image style**.

METHOD



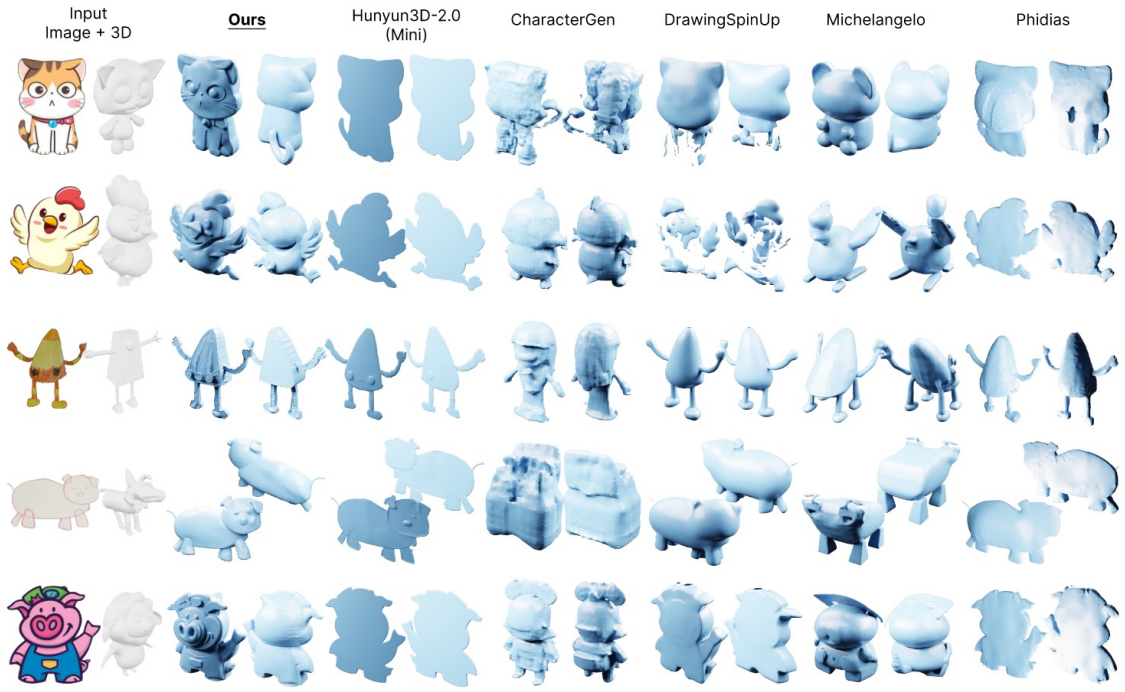
3D Guidance

- Provides **3D guidance from visually similar 3D information** instead of restrictive conditions or image-space transformations.

Visual Editing

- Inspired by FlowEdit, adapts the source 3D by comparing **visual differences between images**.
- **Plug-and-play**, requiring **no training** and applicable to various 3D generative models.

EXPERIMENTS



	Uni3D-I(↑)	ULIP-I(↑)
Hunyuan3D-2.1	0.3039	0.1044
Hunyuan3D-2.0 (Original)	0.2574	0.1020
Hunyuan3D-2.0 (Mini)	0.2384	0.0926
DrawingSpinUp	0.2217	0.0933
CharacterGen	0.2076	0.0945
Phidias	0.2075	0.1352
Michelangelo	0.2784	0.0989
REVIVE3D (Ours)	**0.3537**	**0.1599**

Qualitative Evaluation

- Aligned 3D guidance yields **stable volume**, while FlowEdit-based editing recovers **fine details**.

Quantitative Evaluation

- Achieves superior performance on Uni3D and ULIP image–3D similarity metrics.

PAPER 2

REVIVE 3D: Refinement via Encoded Voluminous Inflated Prior for Volume Enhancement

***Lee, Hankyeol**, Baek, Wooyeol, Kim, Seongdo, and Kim, Jongyoo.*

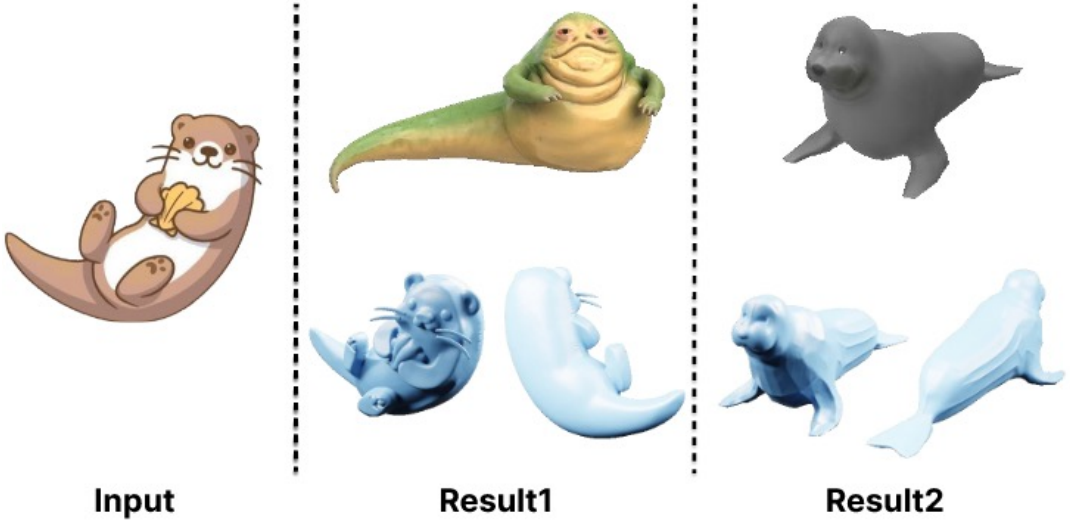
*Accepted at **CVPR 2026 Main conference***

Related Topics

- *3D Generative Model*
- *Latent refinement*
- *Cartoon Image to 3D Generation*

Link: https://arxiv.org/abs/2604.27504

Limitations of Previous Methods





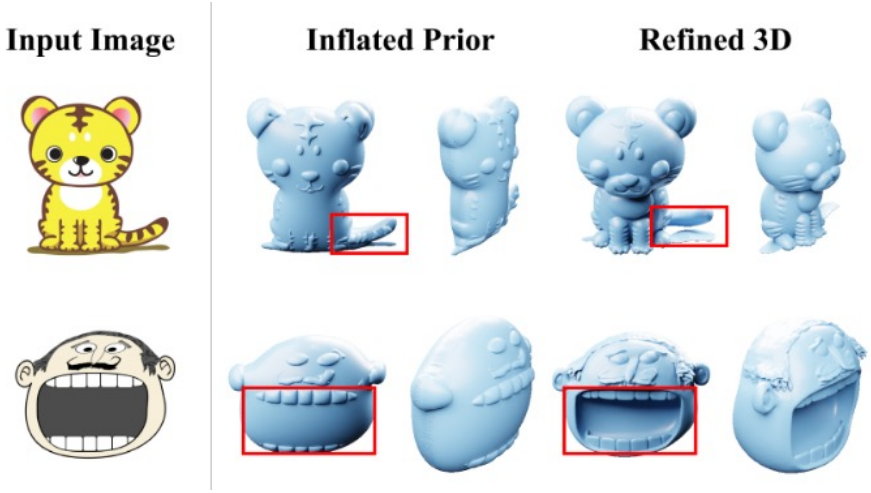
Retrieval Limitation

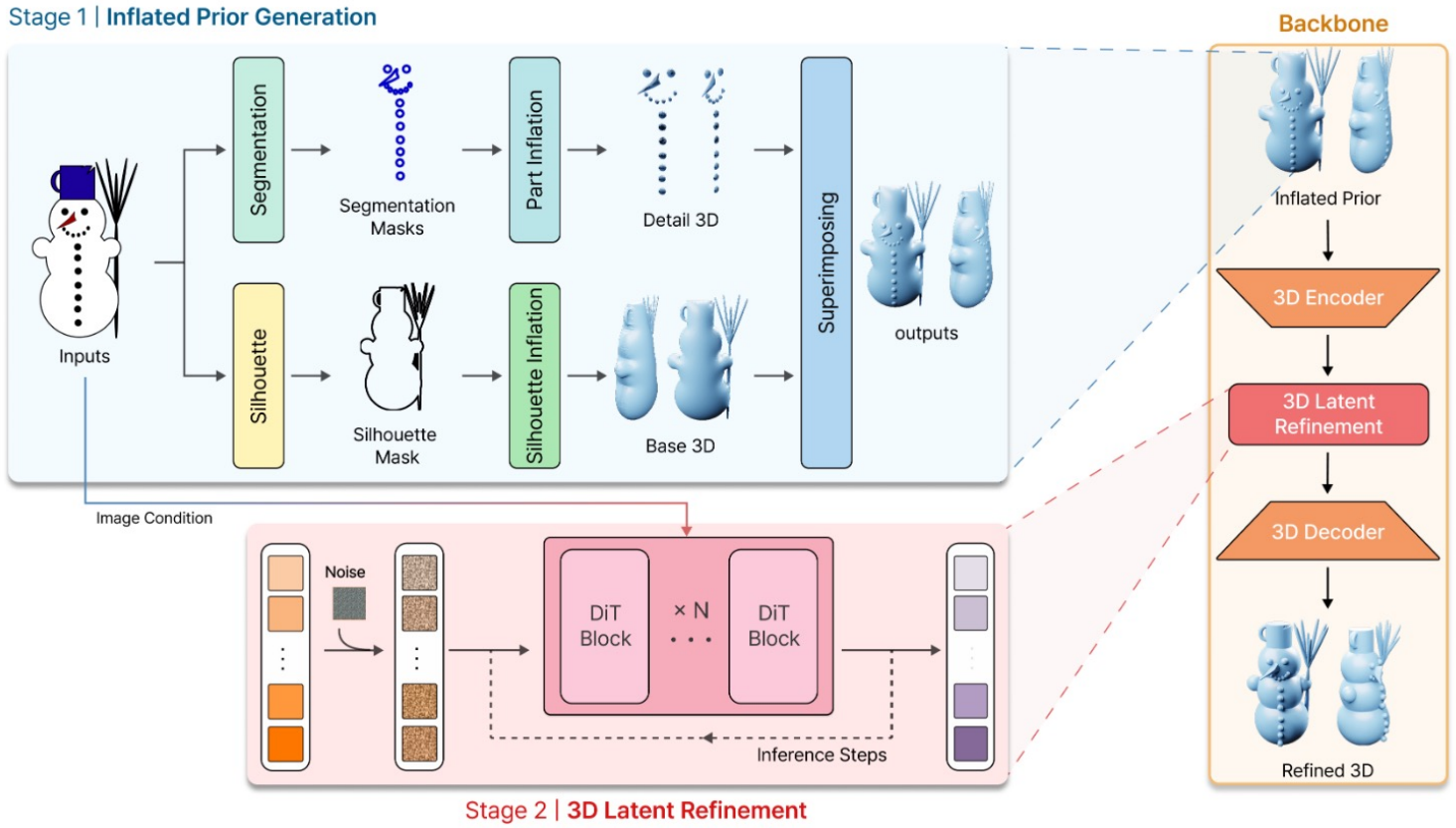
- Semantic-based retrieval often biases the **result toward the retrieved 3D reference** rather than the input image.

Rendering Limitation

- Visual-difference-based updates are **sensitive to rendering viewpoints**.

METHOD





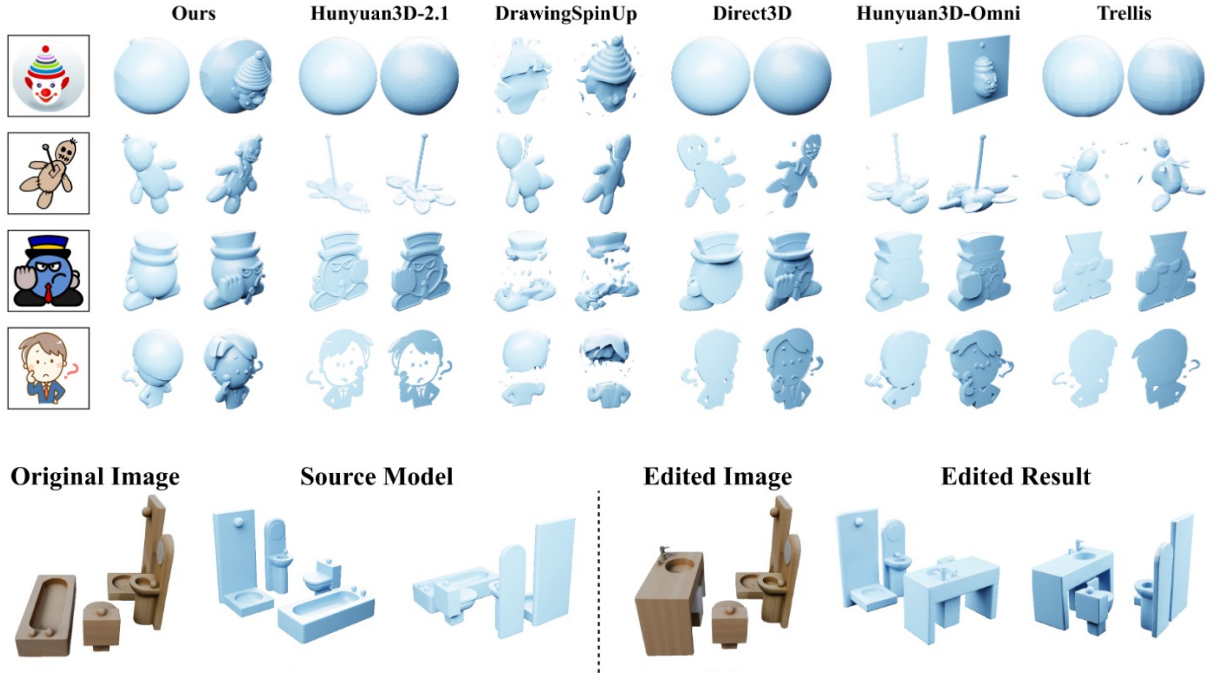
Inflated Prior Generation

- Generates a **3D prior** by **inflating** image contours for global structure and expanded masks for fine details.

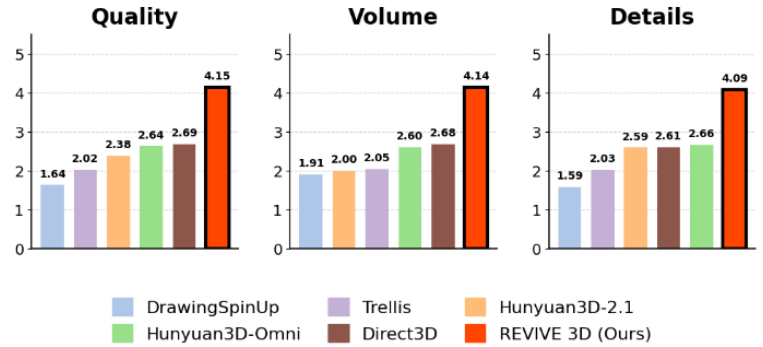
3D Latent Refinement

- Applies **stochastic refinement** from 2D image editing to adapt the 3D prior to the input image.

EXPERIMENTS



Models	Uni3D↑	ULIP↑	C↑	NA↓
Trellis [54]	0.2736	0.1241	0.1748	0.1282
DrawingSpinUp [69]	0.2335	0.1164	0.1604	0.1332
Hunyuan3D-Omni [46]	0.2816	0.1257	0.1707	0.1120
Direct3D [52]	0.2796	0.1315	0.2012	0.1019
Hunyuan3D-2.1 [45]	0.2759	0.1193	0.1408	0.1347
Ours(Hunyuan3D-2.1)	0.3043	0.1265	**0.2179**	**0.0767**
Ours(Direct3D)	**0.3097**	**0.1375**	0.2178	0.0908



Experiments

- **Introduces Compactness (C) and Normal Anisotropy (NA)** to evaluate volumetric quality.
- Achieves **superior performance** across metrics, visual results, and user studies.

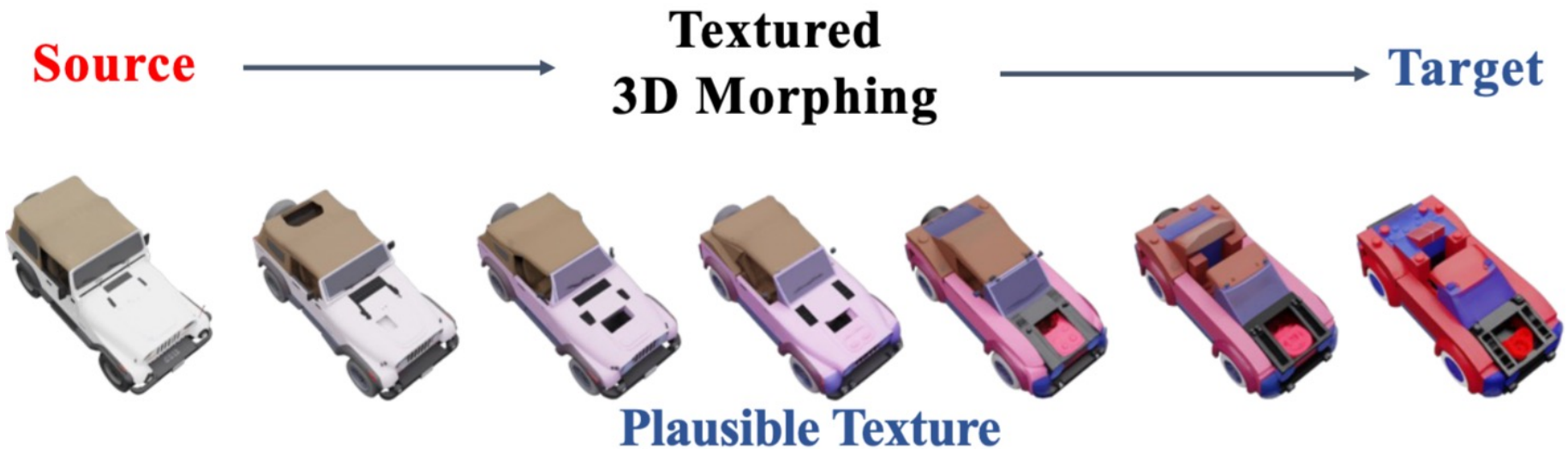
Applications

- **Plug-and-play without training** across 3D pipelines.
- Supports **mask-free 3D editing**.

PROJECT

Image-Free 3D Semantic Morphing

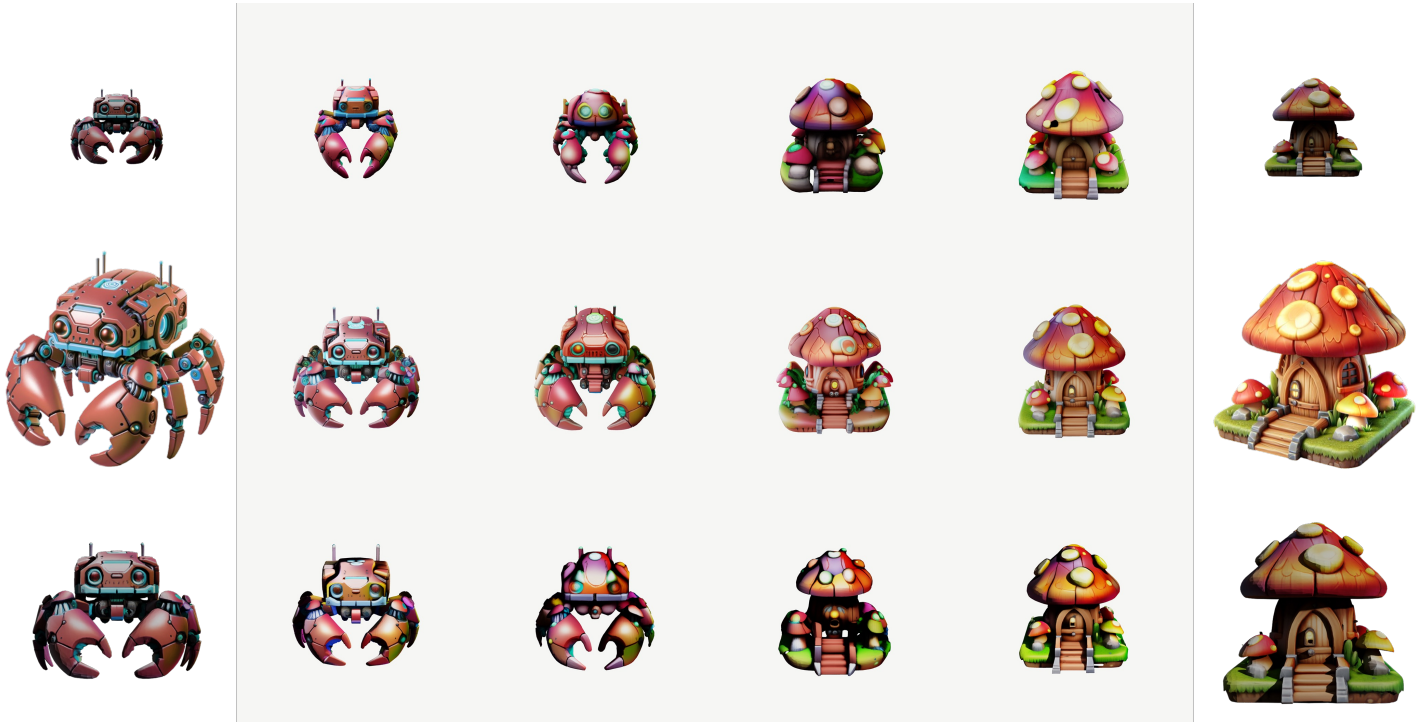
Ongoing Project at NAVER LABS



Related Topics

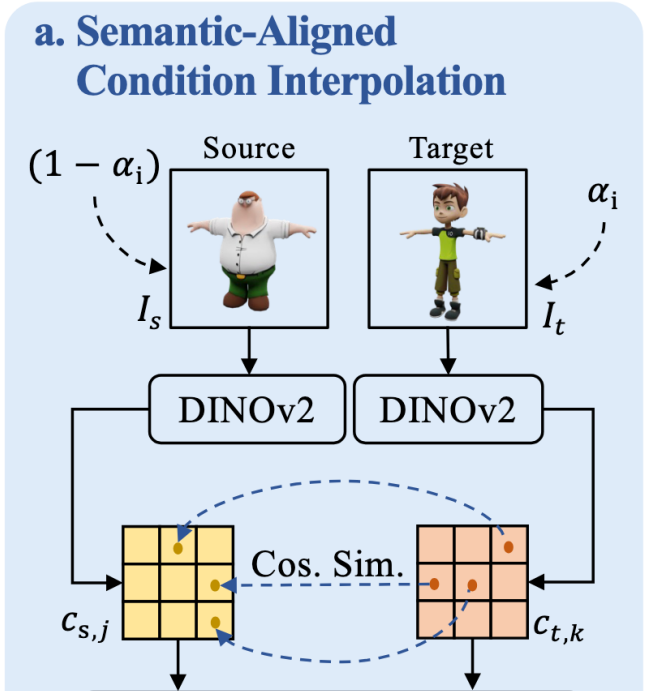
- *3D Generative Model*
- *3D-Conditioned 3D Generation*
- *Semantic 3D Morphing*

MOTIVATION



Source

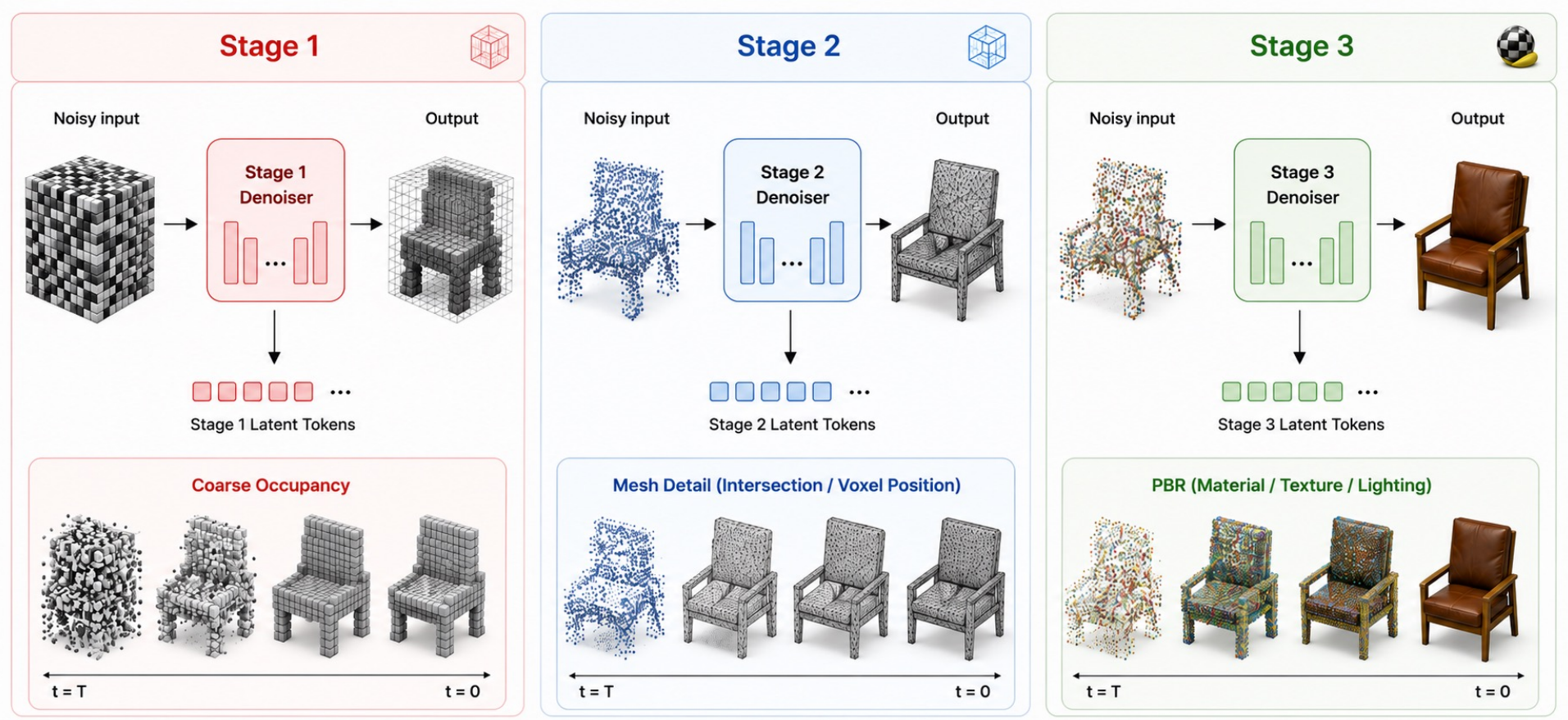
Target



Sensitivity to Rendering Conditions

- 3D morphing relies on rendered-image conditioning, making the results sensitive to rendering conditions.

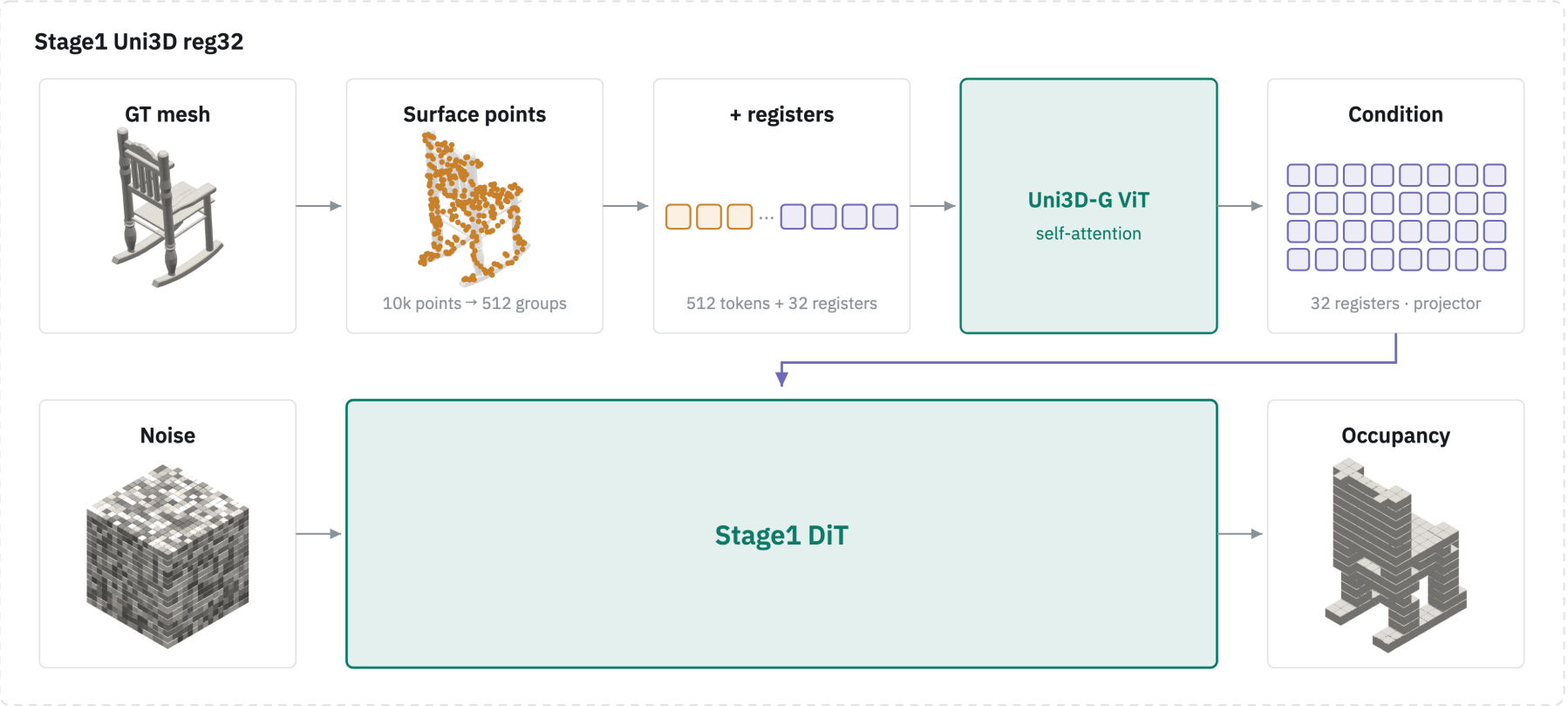
TRELLIS.2



TRELLIS.2: Three-Stage Generation

- Stage 1 — Sparse structure: which voxels are active
- Stage 2 — Geometry: per-voxel shape on those voxels
- Stage 3 — Material: PBR attributes on the same occupancy

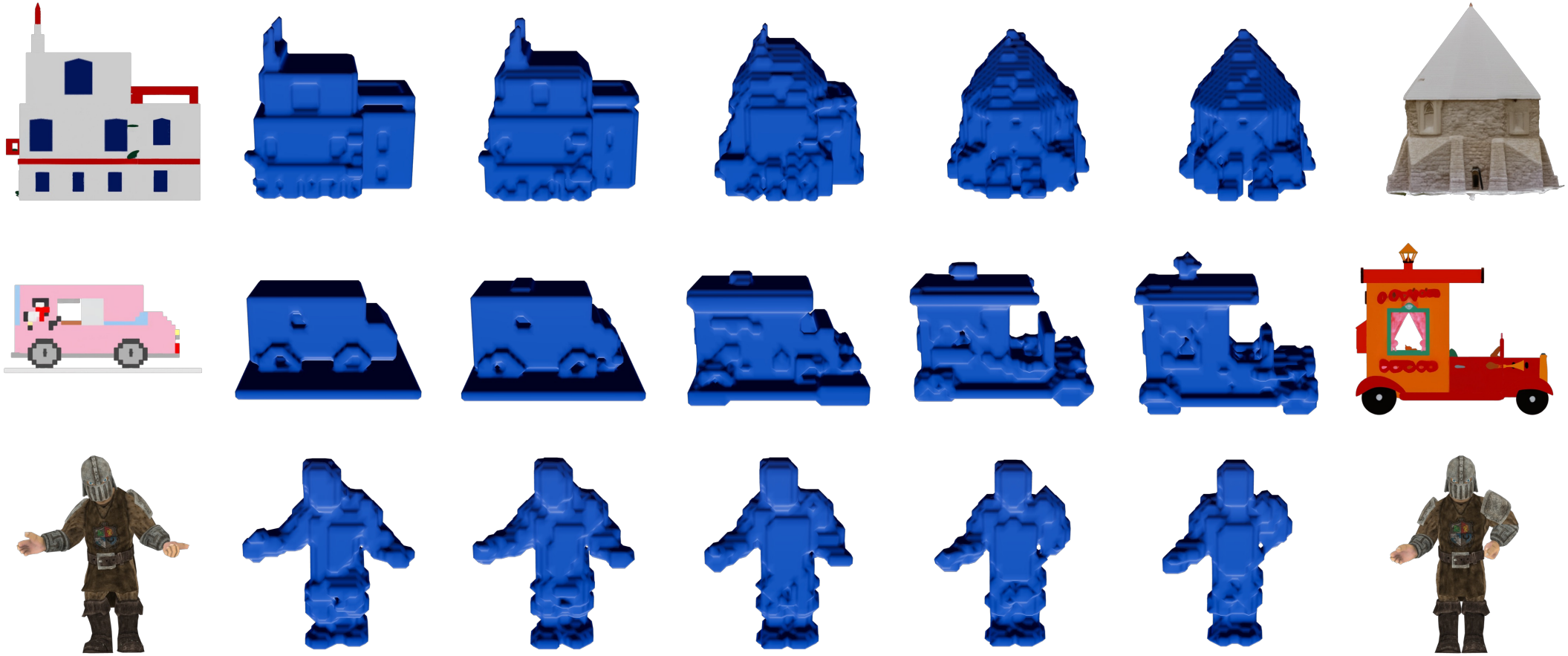
Stage1



Global Register Tokens

- Uni3D embeddings carry position, and blending two meshes blends it, so the intermediates collapse
- Condition switches to 8 register tokens from the Uni3D ViT: no position, global only

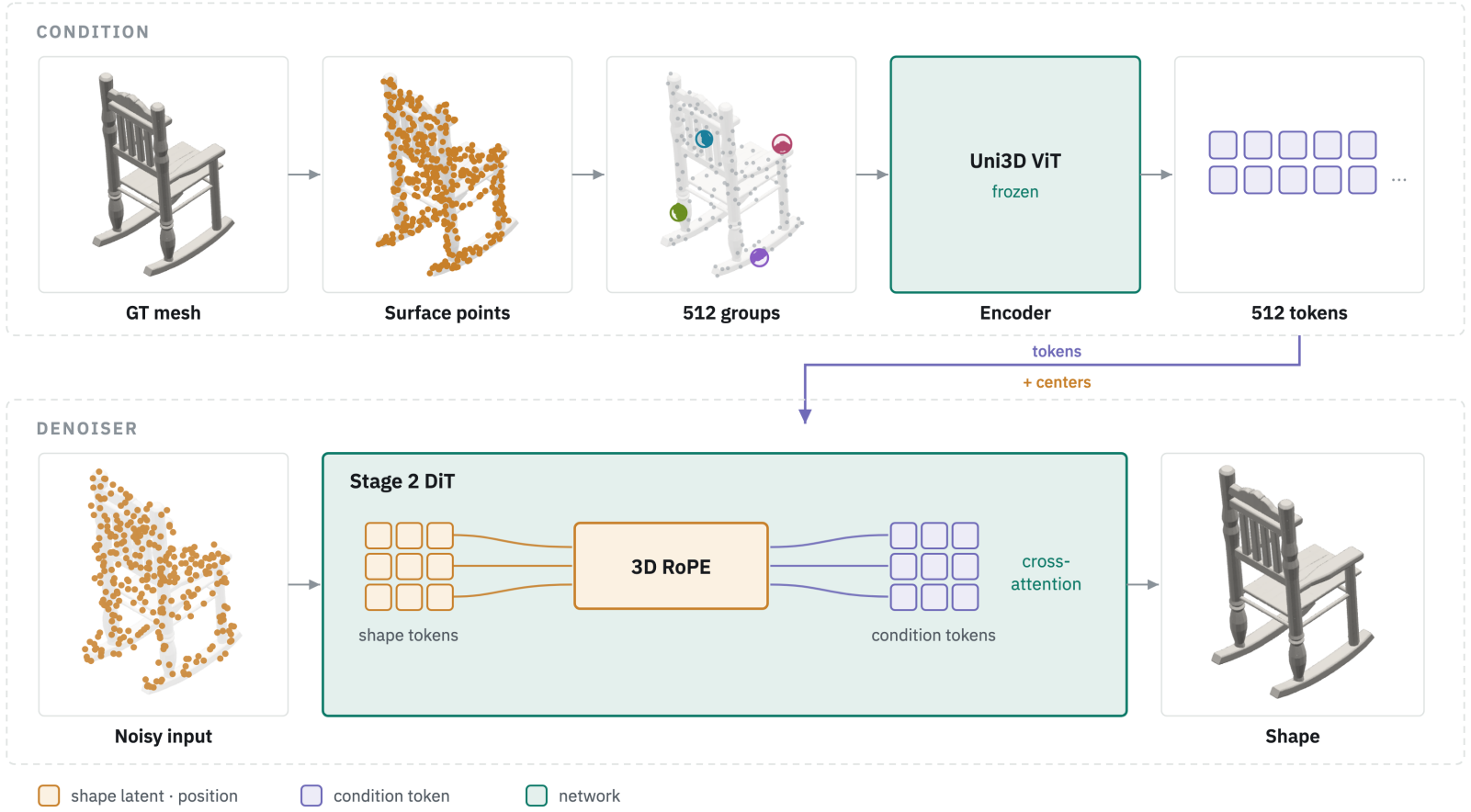
Stage1



Stage 1 Semantic Morphing

- Coarse structure stays one coherent object through the intermediate results
- Register tokens carry no position, so blending them mixes semantics instead of coordinates

Stage2 + 3



Addresses on the Mesh

- Register tokens fail here: detail is filled per voxel, so a token must know which part it belongs to
- 3D RoPE in cross-attention places the Uni3D condition on the mesh

Stage2 + 3



Stage 2 + 3 Semantic Morphing

- Shape and material stay coherent even though the occupancy changes at each step

PAPER 3

Multi-View Cross Attention for Guiding Surgical Planning in Rotator Cuff Repairs With Limited MRI Data

Lee, Yong Oh, **Lee, Hankyeol***, and Kim, Jong-Ho (*Equal contribution)*

*Accepted at **IEEE ACCESS***

Related Topics

- *Medical Image Analysis*
- *Learning under Limited Data*
- *Multi-View Fusion*

Link: https://ieeexplore.ieee.org/document/11303668/

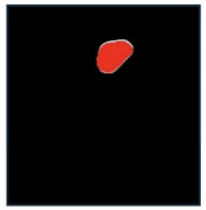
MOTIVATION



Sagittal



[1]Y-shaped Scapula



[2] Supraspinatus muscle



[3] Fat in the supraspinatus fossa



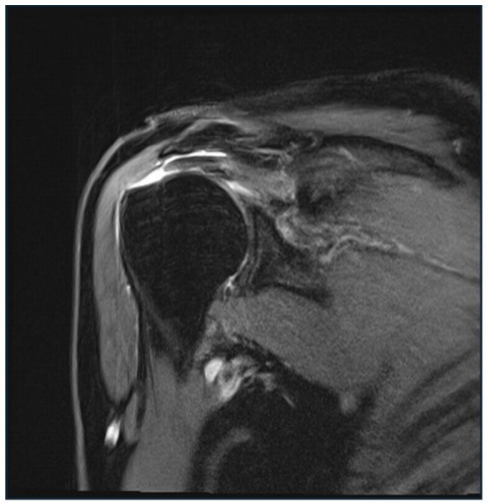
[4] Subscapularis muscle



[5]Deltoid muscle



[6]Clavicle



Coronal



[7]Humeral Head



[8] Supraspinatus tendon & muscle



[9] Glenohumeral joint fluid



[10]Acromion



[11]Glenoid

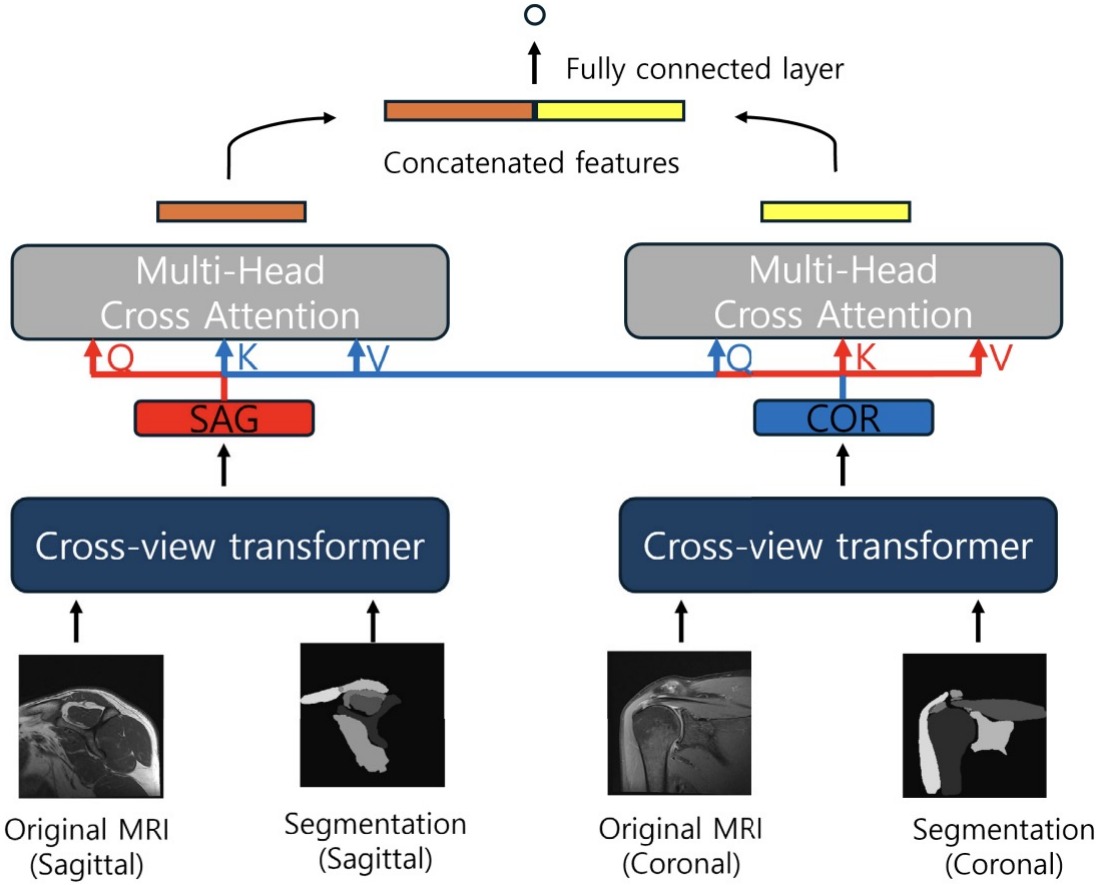


[12] Deltoid muscle

Classification Depends on the Segmentation

- Only 211 MRI pairs, and the target regions are thin and low-contrast, so segmentation stays weak
- Prior work passes those masks straight to the classifier, so weak segmentation drags the classification down

METHOD



The Mask as a Cue, Not an Input

- A Cross-View Transformer lets the mask point to where the classifier should look in the raw MRI
- Extended across views with Multi-Head Cross Attention merging sagittal and coronal
- Mask-only 0.73 → image with mask 0.85 → both views 0.87